SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: ITA 0609 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 40%

QUESTIONS: 15 **Total Marks:25**

Annexure A

Analytical Problem Solving

Code of Conduct:

*I **P.Sai Ganesh Reddy(192525111**)certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
P.Sai Ganesh Reddy

Annexure A

Short Answer Test

1. A simple perceptron model is used to classify whether a student passes or fails based on two input features, namely study hours and attendance. The weights associated with these inputs are 0.6 and 0.4 respectively, and the bias value is –0.5. For a given student with study hours equal to 2 and attendance equal to 1, compute the net input to the perceptron and apply the step activation function to determine the final output. Based on the result, conclude whether the student passes or fails. (5 Marks)

2. In a neural network, a single neuron is trained using the backpropagation learning rule. The initial weight is 0.5 and the learning rate is 0.1. For an input value of 2, the actual output produced by the neuron is 0.6, whereas the target output is 1. Calculate the error in the output and determine the weight update using the delta rule. Hence, compute the new updated weight after one iteration of training. (5 Marks)

3. A deep learning model is developed for disease prediction and its performance is evaluated using a confusion matrix. The model yields 180 true positives, 150 true negatives, 20 false positives, and 50 false negatives. Using these values, compute the accuracy, precision, sensitivity (recall), and specificity of the model. Based on the calculated results, discuss whether the model is reliable for medical diagnosis, considering the importance of minimizing diagnostic errors. (5 Marks)

4. A Genetic Algorithm is applied to maximize the function $f(x) = x^2$ using an initial population consisting of the values 2, 4, 6, and 8. Determine the fitness value for each individual in the population and select the best candidates based on their fitness scores. Perform a crossover operation by exchanging the least significant bits of the selected individuals to generate new offspring. Additionally, explain the role of mutation in improving the diversity and effectiveness of the solution in Genetic Algorithms. (5 Marks)

5 .During the training of a deep neural network model, the training and validation accuracies are observed over multiple epochs. The training accuracy increases steadily from 60% to 95%, while the validation accuracy initially increases from 58% to 72% but then decreases to 68% in later epochs. Analyse this trend and identify the issue affecting the model after a certain point in training. Explain why this issue occurs in neural networks and suggest two effective techniques to address it. Also, discuss the significance of validation accuracy in evaluating the performance of a learning model. (5 Marks)

6. A perceptron model is used to classify whether a loan is approved based on two features: income and credit score. The weights are 0.7 and 0.5, and the bias is –0.6.

For an applicant with income = 1.5 and credit score = 2, compute the net input and apply the step activation function. Determine whether the loan is approved or rejected. (5 Marks)

7. In a neural network, a single neuron uses the delta learning rule. The initial weight is 0.4, learning rate is 0.2, and input is 3. The actual output is 0.5, while the target output is 0.9. Calculate the error, determine the weight update, and compute the new weight after one iteration. (5 Marks)

8. A classification model produces the following confusion matrix values: True Positives = 120, True Negatives = 200, False Positives = 30, and False Negatives = 50. Compute accuracy, precision, recall (sensitivity), specificity, and F1-score. Based on these results, evaluate whether the model is suitable for applications where false negatives are critical. (5 Marks)

9. A Genetic Algorithm is used to maximize the function f(x) = 2x + 3 for an initial population of values {1, 3, 5, 7}. Compute the fitness of each individual and select the top two candidates. Perform a single-point crossover to generate offspring and explain how mutation can help avoid premature convergence. (5 Marks)

10. During training of a neural network, the training loss decreases continuously, while validation loss decreases initially and then starts increasing after several epochs. Identify the issue occurring in the model. Explain the reason behind this behavior and suggest two techniques such as regularization or early stopping to improve generalization performance. (5 Marks)

11. A perceptron model is used to classify whether a customer will purchase a product based on two features: advertisement exposure and income level. The weights are 0.5 and 0.3, and the bias is –0.4. For a customer with advertisement exposure = 2 and income level = 1, compute the net input to the perceptron and apply the step activation function. Determine whether the customer makes a purchase or not. (5 Marks)

12. In a neural network, a neuron is trained using the delta learning rule. The initial weight is 0.6, the learning rate is 0.15, and the input is 2.5. The actual output is 0.7, while the target output is 1. Calculate the error, determine the weight update, and compute the new updated weight after one iteration. (5 Marks)

13. A classification model produces the following confusion matrix values: True Positives = 140, True Negatives = 180, False Positives = 25, and False Negatives = 35. Compute accuracy, precision, recall (sensitivity), specificity, and F1-score. Based on the results, analyze whether the model is suitable for applications such as fraud detection where false positives are costly. (5 Marks)

14. A Genetic Algorithm is applied to maximize the function f(x) = 3x + 2 for an initial population {2, 5, 7, 9}. Compute the fitness value for each individual and select the best candidates. Perform a crossover operation to generate new offspring and explain how mutation helps in maintaining diversity and avoiding local optima.

(5 Marks)

15. During the training of a deep learning model, the training accuracy increases from 65% to 97%, while validation accuracy increases from 63% to 75% and then fluctuates around 72%. Identify the issue present in the model. Explain why this happens and suggest techniques such as dropout and data augmentation to improve the model's performance. Also, discuss the importance of validation metrics in model evaluation. (5 Marks)

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**25**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

1. A simple perceptron model is used to classify whether a student passes or fails based on two input features, namely study hours and attendance. The weights associated with these inputs are 0.6 and 0.4 respectively, and the bias value is –0.5. For a given student with study hours equal to 2 and attendance equal to 1, compute the net input to the perceptron and apply the step activation function to determine the final output. Based on the result, conclude whether the student passes or fails.

(5 Marks**)**

ANSWER:

Given:

- Study hours (x_1) = 2
- Attendance (x_2) = 1
- Weight$_1$ (w_1) = 0.6
- Weight$_2$ (w_2) = 0.4
- Bias (b) = –0.5

Net Input

$$y_{in} = (0.6 \times 2) + (0.4 \times 1) - 0.5$$
$$= 1.2 + 0.4 - 0.5 = 1.1$$

Step Activation Function

- If $y_{in} \geq 0$, Output = 1
- Otherwise, Output = 0

Since 1.1 ≥ 0, Output = 1

Answer: The student passes.

2. In a neural network, a single neuron is trained using the backpropagation learning rule. The initial weight is 0.5 and the learning rate is 0.1. For an input value of 2, the actual output produced by the neuron is 0.6, whereas the target output is 1. Calculate the error in the output and determine the weight update using the delta rule. Hence, compute the new updated weight after one iteration of training.

ANSWER:

Given:

- Initial weight = 0.5
- Learning rate (η) = 0.1
- Input (x) = 2
- Actual output = 0.6
- Target output = 1

Error

$$Error = Target - Actual = 1 - 0.6 = 0.4$$

Weight Update

$$\Delta w = \eta \times Error \times Input$$
$$= 0.1 \times 0.4 \times 2 = 0.08$$

New Weight

$$w_{new} = 0.5 + 0.08 = 0.58$$

Answer:

- Error = 0.4
- Weight update = 0.08
- New weight = 0.58

3. A deep learning model is developed for disease prediction and its performance is evaluated using a confusion matrix. The model yields 180 true positives, 150 true negatives, 20 false positives, and 50 false negatives. Using these values, compute the accuracy, precision, sensitivity (recall), and specificity of the model. Based on the calculated results, discuss whether the model is reliable for medical diagnosis, considering the importance of minimizing diagnostic errors.

ANSWER:
Given:

- TP = 180
- TN = 150
- FP = 20
- FN = 50

Accuracy

$$= \frac{TP + TN}{TP + TN + FP + FN} = \frac{330}{400} = 0.825 = 82.5\%$$

Precision

$$= \frac{180}{180 + 20} = \frac{180}{200} = 90\%$$

Recall (Sensitivity)

$$= \frac{180}{180 + 50} = \frac{180}{230} = 78.26\%$$

Specificity

$$= \frac{150}{150 + 20} = \frac{150}{170} = 88.24\%$$

Answer

- Accuracy = 82.5%
- Precision = 90%
- Recall = 78.26%
- Specificity = 88.24%

Conclusion: The model performs well overall, but the recall is comparatively lower, indicating some disease cases are missed. For medical diagnosis, improving recall would make the model more reliable.

4. During the training of a deep neural network model, the training and validation accuracies are observed over multiple epochs. The training accuracy increases steadily from 60% to 95%, while the validation accuracy initially increases from 58% to 72% but then decreases to 68% in later epochs. Analyse this trend and identify the issue affecting the model after a certain point in training. Explain why this issue occurs in neural networks and suggest two effective techniques to address it. Also, discuss the significance of validation accuracy in evaluating the performance of a learning model.

ANSWER:

Given:

- Training accuracy: 60% → 95%
- Validation accuracy: 58% → 72% → 68%

Issue: Overfitting

Reason

- The model learns the training data too well, including noise.
- It performs poorly on unseen data.

Techniques to Reduce Overfitting

1. Early Stopping
2. Dropout or Regularization (L2)

Importance of Validation Accuracy

- Measures performance on unseen data.
- Helps detect overfitting.
- Assists in selecting the best model.

5. A perceptron model is used to classify whether a loan is approved based on two features: income and credit score. The weights are 0.7 and 0.5, and the bias is –0.6.

For an applicant with income = 1.5 and credit score = 2, compute the net input and apply the step activation function. Determine whether the loan is approved or rejected.

ANSWER:

Given:

- Income (x_1) = 1.5
- Credit Score (x_2) = 2
- Weight_1 (w_1) = 0.7
- Weight_2 (w_2) = 0.5
- Bias (b) = –0.6

Net Input

$$y_{in} = (0.7 \times 1.5) + (0.5 \times 2) - 0.6$$
$$= 1.05 + 1 - 0.6 = 1.45$$

Step Activation Function

- If $y_{in} \geq 0$, Output = 1
- Otherwise, Output = 0

Since 1.45 ≥ 0, Output = 1

Answer: The loan is approved.

6. In a neural network, a single neuron uses the delta learning rule. The initial weight is 0.4, learning rate is 0.2, and input is 3. The actual output is 0.5, while the target output is 0.9.

Calculate the error, determine the weight update, and compute the new weight after one iteration.

ANSWER:

Given:

- Initial weight = 0.4
- Learning rate (η) = 0.2
- Input (x) = 3
- Actual output = 0.5
- Target output = 0.9

Error

$$Error = 0.9 - 0.5 = 0.4$$

Weight Update

$$\Delta w = 0.2 \times 0.4 \times 3 = 0.24$$

New Weight

$$w_{new} = 0.4 + 0.24 = 0.64$$

Answer:

- Error = 0.4
- Weight update = 0.24
- New weight = 0.64

7. A classification model produces the following confusion matrix values: True Positives = 120, True Negatives = 200, False Positives = 30, and False Negatives = 50. Compute accuracy, precision, recall (sensitivity), specificity, and F1-score. Based on these results, evaluate whether the model is suitable for applications where false negatives are critical.

ANSWER:

Given:

- TP = 120
- TN = 200
- FP = 30
- FN = 50

Accuracy

$$= \frac{120 + 200}{120 + 200 + 30 + 50} = \frac{320}{400} = 80\%$$

Precision

$$= \frac{120}{120 + 30} = \frac{120}{150} = 80\%$$

Recall (Sensitivity)

$$= \frac{120}{120 + 50} = \frac{120}{170} = 70.59\%$$

Specificity

$$= \frac{200}{200 + 30} = \frac{200}{230} = 86.96\%$$

F1-Score

$$= \frac{2 \times 0.8 \times 0.7059}{0.8 + 0.7059} \approx 0.75 = 75\%$$

Answer

- Accuracy = 80%
- Precision = 80%
- Recall = 70.59%
- Specificity = 86.96%
- F1-score = 75%

Conclusion: Since recall is only 70.59%, the model misses several positive cases (false negatives). It is not ideal for applications where false negatives are critical.

8. During training of a neural network, the training loss decreases continuously, while validation loss decreases initially and then starts increasing after several epochs. Identify the issue occurring in the model. Explain the reason behind this behavior and suggest two techniques such as regularization or early stopping to improve generalization performance.

ANSWER:

Observation

- Training loss continuously decreases.
- Validation loss decreases initially but later increases.

Issue: Overfitting

Reason

- The model memorizes the training data.
- It cannot generalize well to new data.

Techniques to Improve Generalization

1. Regularization (L1/L2)
2. Early Stopping

Conclusion

These techniques reduce overfitting and improve performance on unseen data.

9. A perceptron model is used to classify whether a customer will purchase a product based on two features: advertisement exposure and income level. The weights are 0.5 and 0.3, and the bias is –0.4. For a customer with advertisement exposure = 2 and income level = 1, compute the net input to the perceptron and apply the step activation function. Determine whether the customer makes a purchase or not.

ANSWER:

Given:

- Advertisement Exposure (x_1) = 2
- Income Level (x_2) = 1
- Weight_1 (w_1) = 0.5
- Weight_2 (w_2) = 0.3
- Bias (b) = –0.4

Net Input

$$y_{in} = (0.5 \times 2) + (0.3 \times 1) - 0.4$$
$$= 1 + 0.3 - 0.4 = 0.9$$

Step Activation Function

- If $y_{in} \geq 0$, Output = 1
- Otherwise, Output = 0

Since $0.9 \geq 0$, Output = 1

Answer: The customer will purchase the product.

10.

In a neural network, a neuron is trained using the delta learning rule. The initial weight is 0.6, the learning rate is 0.15, and the input is 2.5. The actual output is 0.7, while the target output is 1. Calculate the error, determine the weight update, and compute the new updated weight after one iteration.

ANSWER:

Given:

- Initial weight = 0.6
- Learning rate (η) = 0.15
- Input (x) = 2.5
- Actual output = 0.7
- Target output = 1

Error

$$Error = 1 - 0.7 = 0.3$$

Weight Update

$$\Delta w = \eta \times Error \times Input$$
$$= 0.15 \times 0.3 \times 2.5 = 0.1125$$

New Weight

$$w_{new} = 0.6 + 0.1125 = 0.7125$$

Answer:

- Error = 0.3
- Weight update = 0.1125
- New updated weight = 0.7125

11. A classification model produces the following confusion matrix values: True Positives = 140, True Negatives = 180, False Positives = 25, and False Negatives = 35. Compute accuracy, precision, recall (sensitivity), specificity, and F1-score. Based on the results, analyze whether the model is suitable for applications such as fraud detection where false positives are costly.

ANSWER:

Given:

- TP = 140
- TN = 180
- FP = 25
- FN = 35

Accuracy

$$= \frac{140 + 180}{140 + 180 + 25 + 35} = \frac{320}{380} = 84.21\%$$

Precision

$$= \frac{140}{140 + 25} = \frac{140}{165} = 84.85\%$$

Recall (Sensitivity)

$$= \frac{140}{140 + 35} = \frac{140}{175} = 80\%$$

Specificity

$$= \frac{180}{180 + 25} = \frac{180}{205} = 87.80\%$$

F1-Score

$$= \frac{2 \times 0.8485 \times 0.80}{0.8485 + 0.80} \approx 0.824 = 82.4\%$$

Answer:

- Accuracy = 84.21%
- Precision = 84.85%
- Recall = 80%
- Specificity = 87.80%
- F1-score = 82.4%

Conclusion: The model performs well overall. Since false positives are costly in fraud detection, the relatively high precision (84.85%) makes it suitable, though reducing false positives further would improve performance.

12.

During the training of a deep learning model, the training accuracy increases from 65% to 97%, while validation accuracy increases from 63% to 75% and then fluctuates around 72%. Identify the issue present in the model. Explain why this happens and suggest techniques such as dropout and data augmentation to improve the model's performance. Also, discuss the importance of validation metrics in model evaluation.

ANSWER:

Observation

- Training accuracy increases from 65% to 97%.
- Validation accuracy increases from 63% to 75% and then fluctuates around 72%.

Issue: Overfitting

Reason

- The model learns the training data too closely, including noise and specific patterns.
- As a result, it performs worse on unseen validation data.

Techniques to Improve Performance

1. Dropout: Randomly deactivates neurons during training to reduce overfitting.
2. Data Augmentation: Increases the diversity of training data by creating modified versions of existing samples.

Importance of Validation Metrics

- Measure the model's performance on unseen data.
- Help detect overfitting.
- Guide model selection and hyperparameter tuning.
- Ensure the model generalizes well to real-world data.

Conclusion: The model is affected by overfitting. Using dropout, data augmentation, and monitoring validation metrics can improve generalization and overall performance.